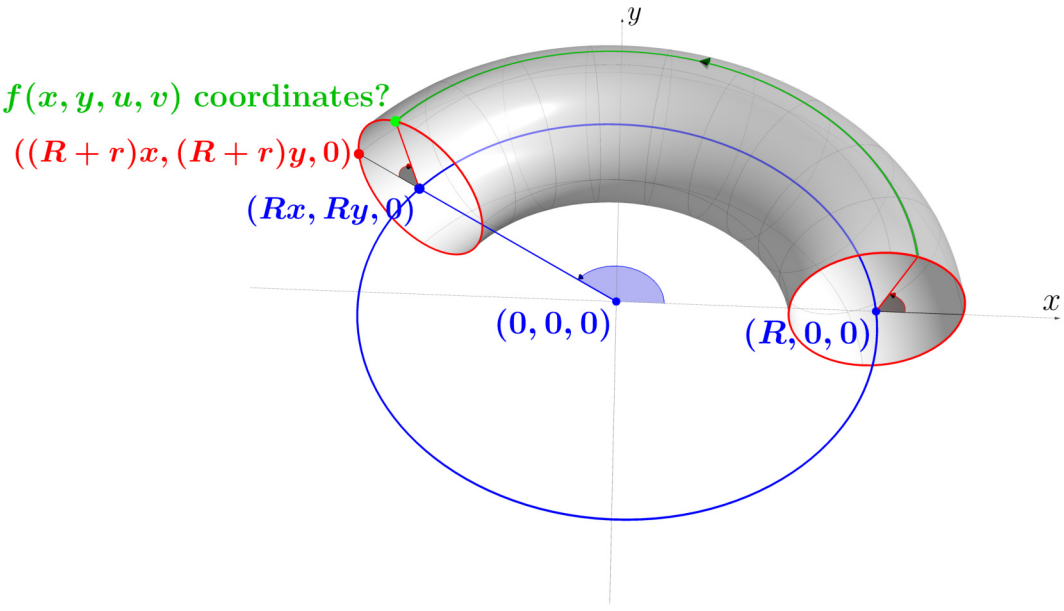

$f(x, y, u, v)$ coordinates?
$((R + r)x, (R + r)y, 0)$
$(Rx, Ry, 0)$
$(0, 0, 0)$
$(R, 0, 0)$
y
x